

Pushkar Dave

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EDUCATION

Northwestern University - Evanston, IL	MS in Robotics (Sep 2024 - Dec 2025)
Visvesvaraya National Institute of Technology - Nagpur, India	BS in Electrical Engineering (2020 - 2024)

SKILLS

Programming: C++, Python, C, MATLAB, C#, Bash, Unit Testing
Robotics: ROS 2, ROS, SLAM, Robot Kinematics, Control Systems, Computer Vision, MuJoCo, MoveIt, OpenCV, Gazebo, RViz
Software: Linux, RTOS, Git, Docker, CMake, PX4, QGroundControl, Unity, Genesis, CoppeliaSim, RobotStudio
Machine Learning: PyTorch, Reinforcement Learning, Deep Learning, Autoencoders, CNNs
Hardware: Onshape, Quadrotors, Quadrupeds, Embedded Systems, Microcontrollers, ESP32, PIC32, UART, SPI, I2C

EXPERIENCE

PSYONIC Inc. San Diego, CA Robotics Applications Engineer	Mar 2026 - Present
- Building a teleoperation system for a robot arm and mounted robotic hand integrating hardware and software components - Leveraged SDKs, open-source libraries, and custom communication bridges to ensure reliable real-time control	
Caterpillar Inc. Decatur, IL Robotics Integration Intern	Jun 2025 - Aug 2025
- Automated visual weld inspection by integrating 3D laser scanners & CV based techniques on production lines - Improved scan quality and resolution by dynamically adjusting sampling rate to welding robot velocity	
Multi-robot Systems Group, Czech Technical University Prague, Czechia Robotics Research Intern	May 2023 - Sept 2023
- Developed a triangulation algorithm in C++ to correct vertical drift for the leader UAV in a swarm system - Performed localization using follower UAVs by fusing UVDAR, IMU, RangeFinder data with a Kalman Filter - Analyzed and recorded ROS simulation metrics and debugged plots to set up real world experiments	
IvLabs, VNIT Nagpur, India Robotics Intern	Jul 2021 - Oct 2021
- Implemented PD control system and minimum snap trajectory generation for quadrotors in MATLAB - Designed a state space quadrotor model and solved seventh-order polynomial functions for trajectories - Experimentally modeled a tethered quadrotor by modeling it as a damped spring-mass system	

PROJECTS

Emergent Locomotion in a Handed Shearing Auxetic (HSA) Actuated Robot	Apr 2025 - Dec 2025
- Modeled HSA actuators in MuJoCo using spring-motor systems to simulate torsional forces driving extension & contraction - Built a Python control interface for the soft robots, enabling stateful rolling and crawling behaviors in simulation - Integrated the Proximal Policy Optimization RL algorithm, to train locomotion models over varied terrain conditions	
Low Level Motor Controller using PIC32	Feb 2025 - Mar 2025
- Designed and implemented a DC motor control system in C using PIC32 with dual interrupt architecture - Programmed firmware featuring PWM signal generation, state machine, encoder communication with UART - Achieved 98% trajectory tracking using PID tuning running on 5KHz current control and 200Hz position control	
Collaborative Mapping using a Quadruped and Quadrotor	Jan 2025 - Mar 2025
- Created occupancy grids using ORB feature extraction, FLANN feature mapping, loop closure using RTABMap in C++ - Optimized ROS 2 middleware, enabled data throttling and transport relay to achieve lossless camera streaming - Deployed multi-session mapping to generate an exhaustive and feature-rich point cloud	
Feedback Control of Omnidirectional Mobile Manipulator	Nov 2024 - Dec 2024
- Generated a cartesian trajectory, simulated kinematics, and implemented feedforward control on a KUKA youBot in Python - Utilized modern screw theory to transform twists into commanded speeds using Jacobian pseudoinverse - Verified the calculations and implementation using ODE physics simulation in CoppeliaSim	